

ActInf GuestStream 056.1 ~ Grégoire Sergeant, "Agency with structured la

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Hello and welcome. It's September 15th, 2023, and we're at the Active Inference Institute in ACTINF Guest Stream number 56.1. Today we have Gregory Sargent-Petri, and we'll be hearing a talk followed by a discussion on Geometry of World Model Influences Behaviors. First, there'll be a presentation, and then any comments that people have in the live chat or any other questions, it'll be great to talk. So thank you a lot for coming. Really looking forward to this. So please, to you. Thank you very much, Daniel, for the invitation. First, I'm really very happy to be able to present this work here. So it's a work around some models of consciousness, at least some computational models of consciousness, some part of consciousness, and how it can help to generate different kind of behaviors, especially when we think about, let's say, artificial agents. So it's a work in collaboration with David Rudroff, Kenneth Williford, Daniel Benquin. In fact, it's not just one work. It's a collection of works that have been on for like already around 10 years. So if you want to know more about like this group and the work that we're doing, you can go on the page on my personal web page, and there's a link to PCM.html. And there's a summary of all the work we've been doing and some summary of the work we've been doing and also of some articles that can be relevant on this subject. So today, we'll focus more particularly on two articles. So one that is accepted and will appear very soon, the other one which we submit, which give the formulation of, let's say, autonomous agents, or like, let's say, mathematical formulation of agents that have a world model that is structured geometrically in such a way that this world model captures some features of consciousness. So the first article is, let's say, the review of the experimental results that we have, and the like most recent formalism that we have on our work, which is the literature on POMD for terrible monitoring process, which is optimal control, stochastic optimal control. And we just try to tweak a bit this formalism to introduce the ideas with how you can include inside of the world model of the agent, some ideas on consciousness, and still continue to have algorithms to do inference to find optimal policies and everything. And the second article focuses more particularly on how changing the world model of the agents can change its behavior, in particular, the relationship, the geometry of the world model, the way that it perceives its environment, and the way it acts with respect to for agent strategy. So the way that it looks for something, so it will change its behavior in looking for something. So I will start by presenting a bit what are all these terms, so world model, what you need it, what it is, what is a foraging strategy, what it is to, how you, how can you define an agent that is looking for a certain object, in particular, what is an exploration based on curiosity, or let's say, more technically epistemic value. So let's consider the setting where you have, let's say, a real world, so the space that's around us, 3D space, you do a setup where you have an agent, which is, let's say, a solid, it has a solid frame. So it has a center, three axis, what is in front, what is on its side, what's above, above it. And it's looking for an object O, which is itself a solid inside of R3.

And all the configuration of the agents and the objects, they are defined by, let's say, a reference frame that is external and that they find that characterizes completely their configuration inside of this world. Now the agent A, what it wants to do is to find for the object O, but it doesn't, it can only have like some noisy observation of where the object O is. So to be able to find O, it needs to have some a priori on where O should be, to be able to plan the consequences of its actions with respect to where O will be once it has moved and to update and to be able to plan what the, how the observation can act with respect to updating its prior. So in another word, so you have, have an agent and the agent is looking for O and thinks that O is, for example, in this direction on the right. So it's going to move towards the right and makes an observation. If it doesn't see O, it's going to change its belief on where O is. So this is a very standard formalism that you can find, like in POMDP, stochastic optimal control. But the one other point we want to, to, to really emphasize is the fact that the agent A has its own frame of reference and the way that it's going to give the coordinates, coordinates of the object, the way that it will trace where the object is, is using its own way of computing the, the coordinates of O. So it has its own way to measure where O is. So it doesn't have to refer to a global frame, a global coordinate system to be able to know where O is. So for example, I am always centered on myself. When I move, I see the object. I don't think about me moving, but I think about the object moving. And this is because when I move, I change my frame. So the way I'm going to reference the object is going to change accordingly. And the point we want to introduce is the fact that you can have several ways of define, for the agent to define its own coordinate system with respect to its environment. And this is a key feature of, uh, that we want to capture, like in terms of modeling, but there's also, we think that it is something that is very structuring with respect to models of consciousness, because when you consider like more generally an agent, uh, that, that had, that is a, let's say that has a world model, uh, in which it has its own, let's say, uh, it has its own point of view perspective on this world model. You kind of had to start tackling the problem of how to introduce a point of view on the environment of the agent. How can you introduce in the way that it's, uh, it is going to, uh, encode its environment, a certain point of view. And in particular, if you think like about it in terms of, uh, the role of space to be able to take into account your own perspective on the environment, you can see that if you take away this, like everything that fills up the space, you keep just the space itself. It's, it's, uh, the space allows you to, uh, let's say to, uh, structure the way that you are going to fill in, uh, some information about the environment and it's centered on your own point of view in a way that when you're going to act, you're still going to keep the still, this, the, the, like the unfilled space, the still, you're still going to keep the stay same way of modeling this space. The space will stay the same when you move, when I move, the space is the same, but I talk into account the fact that I can move from one point to another in this space without changing the space. And something that is also very important is that with respect to any movement that I can do, the space doesn't change. So there's no particular point in space, with respect to the movements that I do. Another way, another property that is very similar is that instead of considering just my actions and

the way that I can change my frame while I'm doing an action, you can also imagine taking the frame of somebody else. So it means that you can, you see that the space that you, you, you use to be able to structure your environment doesn't change when you take the perspective of somebody else. For example, if you imagine yourself as a solid agent changing by a rotation, your own like coordinate, your own solid frame and translating. Okay. So this is, uh, something that we will use in fact, as a definition for the state space, the, the way that the agent encodes its environment. And you will see that there's a natural notion to be able to, that encodes this idea that you can change perspective on your environment and the world model that you have of your environment, simply by introducing something that's very well known, which is the notion of a group acting on a certain space. So this is just to sum up a bit the, what we're trying to do. So we're just trying and trying to do like exploration inside of an environment, but we're adding an additional information, which is that the way the agents is going to encode its environment takes into consideration the fact that it can change its reference frame on this space without singling out any point. Okay. So what do we mean by perspective taking in like more precisely? So how do we go from the real world? So the, let's say the world encode the world described by the external like a frame that allows you to say the agent is here and the object is here from the internal world of the agents. So you just define a map that will take into consideration that the agent can have a first perspective on its environment. The simplest case in the Euclidean case is just rewriting the coordinates of O inside of the frame of the agent, reference frame of the agent. And then once the agent moves its frame, solid frame in the real world changes. And this induces indeed the transformation inside of its internal world, which will be an defined transformation. Okay. But you could imagine also having other transformations. I don't know if you can see my mouse, probably. Can you see my mouse or not? No. Okay. So, yeah. So the map C , can also be something else than an affine transformation. It could be like any kind of transformation. For example, in a fine case, the apparent volume of the object doesn't change. But if you change this map and consider, for example, the projective transformation, as we will describe later, you can have that the size apparent size of the object changes. So we, we, we went from something that is a bit like, uh, already like very well known and not, not very innovative, which is that, uh, you can rewrite, uh, like motions, like the motions depends in the way, it depends on the frame that you choose to describe them into saying that you can in fact change some frames to be able to account for some, uh, like perceptive property. And what we ask also for the agents that we consider is that they can imagine the consequences of their moves, which on their, uh, future observations. So this is something just to say that the agents that we consider have a notion of agency, which is that they can plan the consequences of their actions so that they can choose the best action with respect to a certain reward. In particular, the reward that we consider is the trying to maximize a certain surprise.

So to sum it up, we have an agent that is looking for certain objects. So it has belief of where the subject is in the beliefs lives on, uh, um, um, in an internal state space. So in the world model of the environment, it can make observation, which

allows it to update, uh, its, uh, beliefs, and then it can predict the consequences of its actions so that it can plan the way that it should act with respect to a certain objective. So formally, this is simply the notion of Markov decision process, or more generally like notion of partially observable Markov decision process, because we do not have a complete information of the environment, but also, but only some observations that are, that are limited. And, uh, it's very related. And let's say, I mean, uh, there, there's a strong duality between POMDP and active inference. So there's really a strong link between both. So now what I did in the way that I presented it, so I presented it a bit, the idea, I tried to present the idea of what we're doing. So the general context POMDPs, agents that are planning, uh, their action with respect to a certain objective, here the objective is to find a certain object. And, uh, I give, uh, the formal setting for defining it, which are NDP POMDP. And now we'll, I will go one step further and define it explicitly. And I will continue like this approach for the second part where I will also present our results and our specificity, where else with the general statement and a bit more precise, and then really digging, digging into the results. So here, the classical way to define the Markov decision process is to consider that you have a set of configurations of the environment, which is the state space of the world model in the way, or at least the way that you encode your environment. So this is something, uh, on which you can act because the agent, when it makes

an action, it changes the state of the environment and the actions, the way that the actions change the environment is with respect to a certain, uh, uh, Markov, uh, kernel, uh, probability kernel.

So it's stochastic, which means that each actions changes the state of the environment, but you don't know completely, like it's not deterministic. If you allow some errors in the way that it acts on the environment and you have a reward for a function that is associated to the actions that you can do at time t and also the state of the agent at time t . A partial observable, partially observable Markov decision process is the same thing that a Markov decision process, but you authorize to say that your observation on your environment are only partial, which means that you do not have a precise information on the states of the environment. So the way that you need to relate observation and states is through also a probability kernel, which tells you that, uh, if you're at state, for example, s , you would expect to make a certain observation. Let's say if you, if you think the object is at x , you expect to see the object attacks, but with a certain error, because you know that your sensors are also random. They have some, they have, they can make some mistakes and you keep similarly a reward function. So it's the same thing that an MDP mark of decision process, but you just allow to have some to get information on your environment through observations that are not complete.

And in fact, there's a formulation in terms of partially observable Markov decision processes, our particular case of Markov decision processes, which is called belief Markov and belief MDPs, belief Markov decision process. And the only difference that you have between Markov process and belief MDP

is the fact that the state space is necessarily continuous in the second case. So POMDPs can be seen as a particular case of MDPs. So we said that the actions of a POMDP act on the state space and

you have, you account for the consequences of the actions only through observations. So graphically, it means that you have the state space of yours. So the world model of your environment X on which you, you like you reference, for example, where the object is, so the position of the object.

And then when you act, it induces a consequence on the position of the objects at time t plus one. And then you can make an observation of where you think the object will be, or you can make an observation with respect to where the object is, depending if you're planning into the future, or if you're really doing the action, you're implementing the consequence of the actions in the real world.

So the thing that we're trying to do in our setting is to replace simply the actions by a change of perspective. Like I told you, like when the agent acts, its coordinate system on the environment changes. So you can always see it as in a way passively where you just see it as a change of coordinates.

So we want to say that instead of considering actions on the environment, we have a space on which there's a natural notion of changing of reference frame and actions are simply certain kind of changes of frames. We also allow to have some actions that are, do not correspond to changes of frames. We just say that we include the possibility that some actions are changes of frames and the changes of frames are, have something that is related to all the transformations that are internal of the, for the agents. So for example, things that it knows that a priori, it knows it's encoded in the way the agents will interact with the environment.

So the way that we do it a bit more formally is that we say that changing perspective is simply through the action of the group. So for example, in the affine case, when the, the agent moves, it changes our frame is an affine transformation. And we say that the world model, so the state space is simply a space on which the group acts. So formally, uh, what did, what does it mean that the group acts on the space? It's just saying that, uh, you for, you have space, uh, s , uh, group g , there's an application that goes, that takes an element from g and s and that sends back s . In other words, for every g you have, you can associate a function from s to s and you assume also that it has good properties, which is that, uh, it satisfies equation one. And that, uh, if you don't move, you stay at the same place. So this is like, just the notion of a g space, a space, a group acting on the space. But now if we want to include this in NDPs and POMDPs, we just assume that the state space is a g space, that some actions are, uh, some elements of the group and that when we choose the actions that corresponds to the element of the group, it corresponds to the way that the group acts on the element. So it's very like, uh, there, there's nothing like a very convoluted. It's just saying that you define the collection of functions, which you see as changes of frames and the way that they act. So the property kernel from the state space to the states, for the state space, that time t to time t plus one after the action, uh, after the change of perspective is simply through the, uh, way that the function changes the state space. So it's just kind of a way of reformulation. But what is really hidden behind here is that we have the structure of the group and that we don't consider any kind of actions. We assume that there's more structure in the actions that we can consider. And then it's defined that is, that it's, uh, in like in the way that is, uh, it's it, that it's encoded inside of the geometry of the space that we have. So we don't separate any more action and state space.

We say that the state space in its geometry encodes already certain kinds of actions, which are its change of perspectives. So there are two cases that we consider. The first case is where the state space is the euclidean space and GDF and transformation. It corresponds to the like I find transformations, translations and rotations of the agents and changing, uh, rewriting the coordinates of the objects inside of the ref solid reference frame of the agent. And the second case where, uh, the first, the state space is a projected space and the group is a projected transformation.

You know, the, we described how, what are the generative models that we consider and how we can include. So the, inside of the classical theory of POMDPs, the fact that you can take a perspective inside on your world, uh, on your, on your environment. Now we will introduce, uh, like a classical notion of epistemic value, which will allow us to define what is a behavior, what is an exploratory behavior with respect to curiosity. So how, what it is for an agent to, uh, explore some environment based on curiosity. So curiosity, or let's say the drive for exploration is, uh, uh, uh, quantity that will, it's a, it's a quantity that, uh, you will, um, okay. So it's, uh, okay. It's, so how, how do you define it just with like, very generally. So you start with the agent has a prior on the state of its environments. Then it plans the consequence of one of its move, uh, at the next time and then, uh, uh, at the next time step. So this changes the prior that it has on the environment because there's an action on the environment. So we get a new prior and now it's going to make an observation once. So it, it, it imagines that it's going to make an observation. Once it makes an observation, it's, there's an a posteriori, the, the prior is updated. So you get an a posteriori, the way that you define curiosity or let's say epistemic value is how informative is the observation that you will do. How far is the a posteriori from the a priori, but this, you cannot do it for one given observation because the way that you compute it is by planning what will happen at the next step. So you need to consider all the possible observation that you will make. So the observations are themselves stochastic with respect to the way that you consider, which is that you are a priori on the state space of, uh, uh, at time t .

As we said, for us actions are changes of frame. So we can also define epistemic value for frames, for changes of frames. And in particular, something that we gained that is, I think, interesting is that now we have a function that is defined on the, on the group that can be a continuous group. So you can allow it. You have, for example, if you want to, uh, maximize it, you, you can, or minimize it. And it depends how you see it. You can want, you, you could do gradient descent, for example. So you can have more analytical tools because you're on the space that is continuous and has some structure. So now to give the formal definition of epistemic value. So as I said, uh, it's based on the quantity C , C that I will define now. So if you have yourself, if you give yourself a prior on a space X and you give yourself a probability curve, also stochastic map from X to Y stochastic map is simply saying that for any X , you will associate the measure on Y .

Then you can get a joint distribution on X and Y , which is simply given by the product P_Y doing X times the prior. And in fact, the quantity C is simply the mutual information

between X and Y , which is saying how far is the joint distribution with respect to the, uh, the independent, uh, distribution, uh, the product of, uh, the marginal distribution of X and Y . But this is like, you see it, uh, it appears everywhere, the mutual information. So this formulation is based on the paper that is, uh, from a Friston and all, uh, active influence and the pessimism value. So mutual information is something that appears everywhere, but I think that there's a very nice re-expression of the mutual information, which allows to give a better interpretation of this quantity, which is the input interpretation that I was discussing in the previous slide.

So if you rewrite mutual information, you can always see it as the coolback linear distance between the posterior for a given observation and the prior coolback distance being a way of computing how far you are, how far the two distributions are, but you need to look at the expectation with respect to the observation that you make.

So it's exactly what I was discussing before.

So then, uh, this is like for any kernel, any kernel from X , from Y , X to Y with a prior on X . But as we were discussing here, you have to take into con- so this would be the kernel, but you have to take into consideration that you can do actions and that your prior are on X .

So the way to compute epistemic value in for this kernel here, when having only your, the prior on X is that you propagate the prior by the action on X_1 , then you get a prior on X_1 , and you can define the epistemic value for this prior and a Markov kernel, which responds to the, like the randomness of your sensors, which is always fixed. And this is very important. This one doesn't change. Even if you change frames, the kernel that you have relating your prior, what you think about the state space and relating the state space and the observation never changes. And so explicitly, this means that we, after a certain action or after a certain change of frame, we get the joint distribution on X and Y , which is the following one here. So this corresponds to the prior propagated on X_1 . And then the epistemic value is simply the mutual information of this joint distribution. So now how does the algorithm work? So how the algorithm that corresponds to defining an exploratory behavior for an agent that is looking for it?

So we said, it started with a prior of where O should be. Then you maximize curiosity based on some changes of frames that are around the identity elements, which corresponds to not changing frame. Then you get an action that you can apply, you propagate the prior to the next step with this action. And then you just update your a priori with respect to a certain observation. And then it loops back. So this is the algorithm that we consider in the second paper that I, that was listed in the, in the presentation. So which corresponds to having an agent that is looking for a certain object, his, its behavior is defined by an exploratory, is, is driven by exploration, driven by curiosity, taking into account that its state space is structured by the action of the group. So that the state space is structured by the fact that it, it corresponds to all the possible ways of like changing frames. So it has inside of the state space, inside of the geometry of the state space, all the possible ways of changing frames.

Uh, and, uh, this is, this is like explicitly what we do. So, I mean, in this context, this is the

algorithm that we consider and nothing else. Then we get a very interesting result, at least I find. I mean, it's not, it's, it's interesting in like, in the way that it's presented here, I find it interesting. And then, I mean, if you go more into detail and it's, it's because the Euclidean case is very particular, but what you get is that if you look at the behavior of an agent that is driven by exploration, by curiosity, uh, who, ha the, but that has a state space structured by, uh, occlusion transformation, occlusion transformation. So the first case where the agent is simply encoding its environment in its reference frame, but nothing else, then it doesn't need to move. In the second case where, uh, the, uh, changes of charts are given by projective transformations. So the way it encodes to its transformation, they take into account, uh, uh, perspective, a projective, uh, deformation of its environment, then it will always try to get closer to the object. So you have two very separate behaviors. So this is something that is, I think is, uh, very interesting to, to note. So before going into the details of, uh, how you can prove the statement, I will say a bit more about why it's more, it's an interesting perspective point of view on the, on this subject because, uh, and how it goes a bit further than simply considering this very simple setting. What you could imagine is that encoding the fact that the agent, uh, encoding the actions of the agent directly inside of the state space of the agent through geometry could be a way to stabilize the representations of the agent. And this is something that we're working on now. And there's already, already literature in this direction. Okay. So now let us proof, uh, the statement and to proof the statement, we need to give a formal, like a precise, uh, statement. So what more particularly, what we were able to show is that if you assume that the agent has as move staying still, so it's allowed to stay still, then if, uh, it's, it's, uh, changes of frame are given by a clear, by a fine transformation, the agent stays still. Now in the projective case, if you assume that the agent is always looking in the direction of the object, it will always try to get closer to the object. Okay. So the idea of the proof is that, uh, in the, what, what plays the, the role for discriminant, for, for drive of the agent is how big the agent is appearing to, uh, in the reference, uh, what the, how is the size of the object in the reference frame of the agent. And it's how big it is. It appears to be to the agent. So in the first case, the volume of the, of the object in the internal space of the agent doesn't change. So it doesn't need to move. In the second case, if it makes a move, uh, what is informative is to try to make the object bigger because once it's bigger, the app will be further from the prior with respect to a certain observation. So it will always privilege moves that allow you to make the object look bigger. And in this case, what you can do is, um, uh, you can, uh, you can show that this corresponds in fact to actions or change of frames that gets you closer to the object. Okay. So how do you prove the results? So, as we said, we consider a change of frame from the real world to the internal world. Here, I didn't consider changes of perspectives with respect to moves. Here is simply how you relate the real world to the internal world. In the Euclidean case, it's clear. It's clear. It's just that the way that you write the coordinates of the object in the solid frame in the projective case, it's clearly not, it's not obvious because there are several ways that you can relate the side frame of the

agent to the projective frame. So one way that we decided to do it is we give some set of axioms that we consider to be, let's say, uh, coherent with our own experience of, uh, of space, which is that we feel that we're always centered on ourself, that the axis of the side frame of the agent inside of the, uh, like three space. So what is in front of it, or what is on the right, what is top are preserved, that there's no point in front of the object that appears to be at infinity and that that near to the center of the object, the volumes are preserved.

So when you do this, so this, uh, this formulation, like this, uh, result is in, in this article here.

So the way that we, uh, relate some, uh, solid frames, like some, uh, solid, uh, reference frame of the agent to some, uh, projective transformations are in this article, in particular in proposition A1, where we show that this set of axioms limits the set of projective transformation we can consider, and we will only have this projective transformation. And so the change of frames from the external world to the internal world is given by rewriting the coordinates of the agent, of the object in the solid frame of the agent, and then applying this projective transformation.

So now, uh, what we defined here is, uh, are the maps that relate the observation of the agents.

So the, we, we defined the maps in the Euclidean and projective case that relate the observation of the agent,

the observation of the object to the way that it represents this, this object inside of its internal world.

So here, so this is simply like the POMDP that we defined before. And now once the old agent doesn't move, it changes its Euclidean reference frame. So there is a solid reference frame. So that from going from one side reference frame to another side reference frame, you can, by applying the projective transformation we had before, define another one that we call ψ , that goes from the state space at time zero to the state space after action, after moving. Very, what is very important is that we consider, uh, the Markov kernel associated to a noisy observation to be of this shape here, which is that if you know that the object, if you think the object is at point X , then you believe that the observation will be around X , uh, in the ball, uh, around X for a ball of a certain radius, which is a small radius, ϵ . Okay. And I need to charge my computer. So sorry.

Okay. So now press, uh, epistemic value as defined. Uh, uh, uh, curve, uh, uh, uh, within the, uh, from my limit.

With, oh, in fact, that it has a very simple expression.

So the epistemic value of the prior propagated after the transformation,

Φ after this changes of frame, Φ . So you have a prior here, you propagate it on X one through Φ , and you compute the epistemic value for this, uh, joint, for the joint distribution over X one and Y is given by this formula here. Well, simply integrating overall the possible places where the object could be times the probabilistic volume of, uh, the ball, um, of size ϵ , uh, for the propagated, uh, measure and, uh, logarithmic of the same quantity.

Okay. So this is very direct to right. You can just, uh, compute it and you, you find this expression, uh, but what you have already here is that if you consider an occlusion transformation, then the quantity Q side minus one $B \epsilon Y$ doesn't change. So then you have the epistemic

value that is a, that is constant. So basically if you try to maximize epistemic value, maximize this quantity, you can do anything. You have always one move, which is not, which is not you move. So you don't move. It's okay. It's perfectly fine. Now, in the second case, we have the projective, where you consider that, uh, the way you relate environment and the internal world is through a projective transformation, then it's more complicated. And so you need to, uh, use a trick, which is that, you know, that once an observation has been made, the support of the prior will be smaller. And so after one step, if you suppose that your epsilon, so the size of, uh, let's say noisiness of your sensor is small enough, uh, you have a support that is, uh, of the distribution that's sufficient, uh, small enough so that you can do an asymptotic development of the quantity in the internal. So this is an equality, but this is an approximation here.

And now this is very useful because now that you know, we're like doing this in a way, uh, you can, uh, say that you just need to develop this quantity at the point where the object is really.

So if you do it, you get this expression here. And what appears to play a role is only the determinant of, uh, the Jacobian or the projective transformation, which here in the case will be one, when in projective case can be many things. So here also, it's not very clear because how do you define CM? So CM is defined as a composition of several maps. I didn't go into the details, but it's corresponds to the map that are given by the changes of frame. So if you have the Euclidean space and the internal spaces, after moving, you have a change of frame in the Euclidean space, but it corresponds to also a change of frame in the internal space. So if you, the way you define ψ is simply saying that you inverse the projective transformation from internal world to external world to go at time zero, then you apply the change of frames and then you apply the projective transformation to go from external to internal. And so it's exactly this formula that you have here, and which is very nice with this. And then you have the first term that doesn't depend on the actions that you do. So you don't need to take into account. This one will be one. And then you just need to compute this one. In fact, this one has very simple expression. So it's expression 23. And then here you can directly see which moves correspond to increasing this quantity and decreasing this quantity because we want to increase epistemic value. And so this, this is the results. This allows to prove the result that we said in the first day, like in the several slides before that in the projective case, if you have enough movements, if you look at the object, then you're always going to go closer to the object. The way to interpret it is as if the agent, it was a bit paranoiac or paranoid, a bit like he's very uncertain on its own beliefs. So it knows the object might be over there, but it's always uncertain. So it needs to go check. And once it's checked, it's more certain, but still, as it can always be even more certain, it will always try to get more and more certainty.

And so I only presented like in this, in this presentation, all the aspects which are more computational and the algorithm and some like analysis of the algorithm we considered. But we also have like some experiments on how this setting can allow to generate different behaviors, behaviors that we would expect, explain, for example, some illusions like the moon illusions. And I invite you if you're interested in this to listen on to the like online talk of like the MOC4 conference that

was in Oxford last week, or to check out one of these three papers. And I would like to thank you very much for your attention.

All right, awesome. Wow. Very interesting and different ways than how we've seen the POMDP and related works. So, okay, let's just start off with a little context, and then I'll read some questions and read some questions from live chat. So what brought you to study this question this way? What question? The question of, okay. Did you come from a math side and find consciousness and geometries to be interesting or vice versa? What kind of brought you to want to make this contribution? So at the beginning, so when I did my PhD in maths, I was more interested, I was interested in, let's say, I call this idea of critical brain hypothesis. So the way that the trying to understand the way that the brain processes information and makes it something that is, can be like exploited. And so they, one of the, so the critical brain hypothesis tells you that the activity of the neurons is basically close to a certain criticality, criticality in the source of statistical physics, if you, because you can model the activation of the neurons as a, let's say a statistical system, like an Ising model. But the, so I worked a lot on this. Then I, there's another hypothesis that is like very common, which is the Bayesian brain hypothesis. And so the Bayesian brain hypothesis led me to active inference, to learning more about optimal control Bayesian, Bayesian perspective on optimal control. And, uh, there was a, so, uh, my advisor was a PhD advisor was working with David and they still work together on, uh, uh, trying to, uh, implement some aspects of consciousness, uh, and how it can influence inference, especially with their article in the moon illusion. So, uh, I w I was interested in knowing how, uh, how, uh, this, uh, kind of, uh, let's say, uh, ideas, uh, simply like in a very naive way, they interact with the Bayesian brain hypothesis. And then I continued in fact, like one of the lines of my research is, uh, structured, like algebraically structured, uh, statistics or machine learning. And so the way that they see it like very geometrically, in fact, or let's say geometrically or algebraic, it's not the same thing. It's very related. It's something that I, I wanted to, uh, to, to, uh, understand a bit better, let's say to, to make it in a formal setting so that, uh, it's simply a particular case of what are, what is in the literature. So it's specifying what exists in the literature. And this took some time because, uh, I was more on the active inference, like the free energy principle side. So I had to read more about like, uh, optimal control, so classic optimal control, POMDPs, and understand that, uh, if that, what we're doing is simply adding more structure on the latent space of, uh, of, uh, of the agent and the state space of the agent and, uh, doing this allows to ask the question, like, why is it useful? Why having a state space that encodes different perspectives? So the motivation comes from the consciousness study, cognitive sciences, but why it can be useful for robotics. And it's always been like my motivation study, uh, some, uh, statistical models, more structured models with more a priori that can be, uh, useful, uh, for understanding the behavior of a closed system, like an agent, like a collection of, uh, neurons, like even like, um, molecular machines. Okay. So it's a very long answer, but it's, uh, okay. Cool. So you focused on the spatial movement, epistemic foraging case. Is there something special

about space or can we also think about this perspective taking in terms of, for example, semantic or a narrative reference frame? So the, this is, I think it's a very, very important question because up to now, uh, all the work, a reference only to space. So the thing is space in terms of a 3d space and not space in terms of geometry. And the fact of writing it as geometry allows to get out from the classical point of view, a space as a 3d space, because you see that there's more and more like, for example, in the, in, uh, geometric deep learning, there's more and more the use of, uh, space to encode in variants of certain objects. And these objects are not necessarily, uh, have the three, the three dimensional structures. You can have objects that have higher groups of environments. So I think that it's, it's indeed using geometry for other contexts. And it's really the aim of trying to go to this more general formulation is to be able to apply it to real world models. What I mean by real world models, it means the ones that are learned to be able to have an agent in an open environment that will learn the way, uh, it would learn its generative model, but with the app priori that it can take a perspective on its environment and see what it can do. So we really want to get out from the 3d, at least for me, go out from 3d space and go to, uh, like implementing it completely, uh, in autonomous agents.

Okay. To kind of follow on that. There's a question in the chat chat. Great talk. I also wonder whether it applies to any modality, not just visual spatial, but also text.

Yeah. So it's, it's for other modalities like sound, for example, but I don't know for applying to text there. So there's a lot of work now on, uh, large language models and like, uh, of this idea of prompting and being, having these models, being able to have some kind of imagination. So you would like to have this kind of, these ideas applied in this context, but for now, I mean, it's not the line that we're trying to, that I'm trying to do. So I'm trying to stay on deep learning. So basically take standard data set without considering, uh, uh, uh, like texts and just try to see how these ideas like in geometric deep learning can stabilize representations. So it's not the same that is in others. It's clearly other modalities, especially if we consider like a multimodal integration, like you, you try to rebuild the state space and you don't want to see it only as a vector space, but you wanted to see it with more structure because you want to force the way like you have constraints be, you know, a priori the constraints that you're going to have on the way that you can take a perspective on one modality or the other, but it's not for text. At least not, but I think it maybe it could be used for text, but, but it's not what I'm doing now. Can I have something to say about this question?

Yeah.

Sorry. I just want to rebound on that.

I have the luck of collaborating with a real mathematician.

I'm just some guy with intuition that I found the right people to do the work.

Regarding multimodality, this is important to understand that this is not about vision.

This is about spatial cognition. It's supramodal. The claim is that vision is just one particular way of integrating information indeed in an obviously projective manner,

but that's integrated in a much larger field of experience than the field of view or the visual field and obviously proprioception touch when you build a representation just by touching something and you get this 3D representation of that thing automatically in your mind, hearing to the extent that it's about source localization and building spatial representation. All of that stuff, that's the claim of the theory, is integrated in this projective space. There are priors from memories, there are stuff from visions, stuff from audition, stuff from proprioception, interoception, you name it. All senses contribute to it. So it's not vision. The claim is that 3D projective geometry in that case is beyond vision. Vision is just actually a slave to that supramodal representation. That's the claim. Awesome. Yeah, please.

That's important because we do, I think we, like we, so we work like a group. There are several, like different perspectives for me. I'm more on the computational side, so it's important to have that color because clearly something I won't be able to answer. Cool. Here's a nice following question from Vladimir in the chat. They wrote, if the e.g. visual sensors have variable resolution, for example, higher resolution in the center, this can naturally lead to curiosity-based change of orientation response within your framework. So is it the question or an affirmation? So it's for e.g., right? So it's... If there's variable, if there's variable sensor precision, for example, higher precision in the center, might you see any resulting curiosity-associated change merely based upon the asymmetry or the structure of the sensor fields? Okay. So I shouldn't say it, but I will still say it because I don't know if the paper will go out one day or another. This is more attention. This is more attention than curiosity. So it's more like, so you can act on your sensor so that you can change the way that you integrate them. And this really acts as a form of attention. So I really went into it.

What do you think that... What is the interplay between attention and curiosity?

Uh... I mean...

It's the way you're going to move, basically. On the... The way... The way that... It deforms the way that you move.

And you... So the consequences of your actions are not the same. So basically curiosity is what...

What drives actions? And the way that you... So you choose your action with respect to reward, which is epistemic value. And now attention, or let's say changing the sensors, is a way to change the consequences of your action with respect to, let's say, optimizing this value. So it's like...

Um... Yeah. I mean, it's... It's... Yeah. I... I don't know if I should say yes or not. But okay. Okay.

Let's say it's okay. That's it. I think it's a metric, basically. It's... It acts like a sort of metric on the space, on the... The group directly. So it means that it's going to... So when you have the function, you want to do gradient descent, you need to choose a metric. And the metric is... So the attention will act as a metric.

And so the metric will allow you to deform, in fact, the... The steps that you will do. And so I think this is how it acts.

So it's very known that... I mean, it's known that the fact that changing the sensors is... Is related to attention. That's not something much new. But the fact that you can directly, in our setting, relate it to groups.

And you can relate it to... Like changing the metric on the group is something that... That can be done.

And I think it's... You know, it's interesting.

All right. Another question in the chat.

What is a formal framework for learning geometry from data?

How do we move from empirical data sets, the files on our computers, and the things that we do deep learning on, take into machine learning pipelines? And how do we utilize geometric approaches and formalize to the kind of analytical precision that we saw here, some of these geometric relationships?

So there's a very like... There are several different fields of how to use geometry on data. So there's already like... I mean, depends how you see geometry. But one way is a TDA, topological data analysis.

So which is basically trying to... Let's say... Do you want to provide learning by geometric...

Like stability property of your data? And try to interpret it as a certain kind of space. And

then you look at the whole of your space. And this is something that defines your data set. So this is one approach. Another one is that people doing... Like they try to learn the... So they know that you're...

That the... That the data lives in the low dimensional manifold. And they try to learn this manifold.

So that's another perspective... Another way of doing it. So that's a lot of work in this direction.

There are people who are interested in variance and equivariance. So more in the geometric deep learning.

There are people who are interested in the same setting on a priori. How to use the geometric a priori and included in learning for deep learning reinforcement learning. And this is what we do. In fact, this is one of... Many people do this. Not just us. But we like for us, we try to focus on this idea of how to exploit it for reinforcement learning. Hmm. So in this setting, epistemic value was the only driver of action? Is that...? Yeah.

So it's kind of like expected free energy, except without pragmatic value. So we only have the epistemic term remaining.

Exactly. Exactly. In fact, if you look at it in terms of optimal control and not by Asian,

because you know that there's this duality that is most of the time stated in like active inference,

where you have basically a duality between the, let's say, like stochastic value function and the probabilistic version of value function. So you can encode like on priors, you can encode all your drives with priors or with the value function directly. So rewards. And so both are kind of dual and they're dual in a way. You can make like this duality. I mean, it's... I don't know if it's explicitly dual.

Like in some contexts, this sentence makes sense. For active inference, for me, I'm not completely aware if the duality is exactly formal, but at least the idea is here. So there's no big difference,

at least the way I see it as with value function and seeing it probabilistically. But the terms of

epistemic value is an exploration drive, which you also find in reinforcement learning,

maybe not in this exact expression. I think that the exact expression that was given in the paper of Frissan and Hall is... I think it's Pezzoulo and Onibélé. Onibélé for sure, I know it is on the paper.

It's very canonical way to define epistemic value. So it's an exploratory drive. You can always add to the value function.

You know, but there's a lot of problem of trying to explore. In fact, like your environment,

you find the good policies. Because if you're in a state space that is continuous, it's really difficult to result to result in P1EP.

Yes. It's like if you knew which curiosities you could get rewarded from, you would have already known the answer to the search. So that's one of the challenges with pragmatic value. It converges well to expectations, but then this work really focuses in on epistemic value and shows what it can do alone as a driver. So how would you bring pragmatic value into this formalism?

You just added. It's really... You can put the value function, so some of rewards in some way, and then you add the epistemic value. In fact, when you look at the formulation in terms of belief MDP for POMDP, the curiosity epistemic value is simply a value function, nothing more. It's a one-step value function. But I mean, this is like just playing with definitions. But so you can always add a drive for exploration. And this is something that's really often done in reinforcement learning. I mean, like...

I think it's even standard in... Not this way, not exactly this expression, but there's a book that's called Reinforcement Learning State of the Art, and they introduced this exploratory drive. Like this book maybe that has 10 years or something like this.

So it's something that you add up. You can put... The way we will do it now is like we can put some drives with respect to preferences, and then you add, for example, an epistemic value. And you try to just solve the optimization problem. But in terms of formalism, if you look at the belief MDP, it's nothing more than a certain value function. So it's not...

So space remains when we translate through it, or when we...
Yeah.

So space is basically the... What is not changed through action. Is that the case? Okay.

Yeah. And so...

It's just the way... So the thing is that... So... Okay.

So there's a bit of technicality. So the way that we use is we use a chart, so we take a...

For the projective plane. So we need to use homogeneous coordinates. So we need to take away like a lower dimensional plane. So that's why we always have the same like in the way that we encoded, we always are in R^3 . Because we chose a chart. I mean, we chose homogeneous coordinates. And we did projective transformations from one homogeneous coordinate to another homogeneous coordinate, but it always lifts to a projective transformation. So what is hidden in the way that we write it is, in fact, we don't have the same space. In the first case, we have Euclidean. In the second, we have projective. So... But the way as we use charts, and we like... I mean, we don't put all the details of the fact that, for example, you can compose two projective transformations in...

Even if you write it in charts in terms of homogeneous coordinates, and it stays a projective transformation, this is something that is okay, but it's... We just don't write it this way. We stay in charts. And so that's why there's a similarity between... Because we want to implement it. There's a similarity between the Euclidean... Euclidean space... Projective case. Euclidean

case and projective case have the same state space R^3 . But the space of transformation is really different. And in fact, they are not the same space. But what is important for us in terms of space is just to tell yourself that if you take away everything that's inside the space, all the... Like everything that populates space, you take it away. What you're left with is with this kind of concepts, which already takes into account the fact that when you act, you're not changing... You're not changing the space. And there's no single point that is singled out. So this is very important because it's a very typical... Like it types a lot of the object you're looking at. If you move, there's no point in the space that is changing. There's no point that is singled out. And when you move, the space is not changing. So you know that you...

You already have inside of your space a change of charts that is encoded. And even more when you can imagine that you take the perspective of somebody else. And this is like really something that is at the basis of the space we consider is that the space is simply a way to support the fact that you can change charts, you can change perspectives. And so it's including this idea inside of agency that we're trying to do. So maybe I was not clear in the way that I did it first, but I think this is really... What do you do?

What do you do?

It's the water we're in. So it's a very interesting way to approach it. And it reminds me way back when, actually almost three years ago, when we discussed the projective consciousness model and phenomenal selfhood in live stream number nine. And we talked a lot about flipping between the Euclidean and the projective modes, how an agent could have on one hand, a space in which a book is a rectangle, and yet also be seeing it very close to their face so that its visual projection was different. And yet here, there was different behavior associated with the frame alone. So what's going on there where even though they apparently can be reconstructed from each other, what flips can we flip from thinking more Euclidean and then in that situation, um, you know, I'm a plane 30,000 feet above my city, so there's nowhere I need to go. But then in the projective setting, we do have this kind of inbuilt epistemic drive to be near.

So I think it's a hard code. So the idea is, one of the things that it can be useful for is communication. So using space as a way to communicate, taking on as a priori that we have very different architectures, like the way that we treat information are not the same, we don't have exactly the same connection, the loans and everything. But we still have like a common framework in which we can discuss, for example, visual information is something that is very immediate for us, which is not obvious if you take the perspective, the point of view, if you go from the idea that it's rebuilt, and it's not the environment you're living on, it's not in, it's not exactly the space that we see, there's something that we constructed like, for like, functional reasons, and that is one of them could be that we can communicate very directly with it. So I think it's hard coded that the state space needs to have this action. And that all the agents share this action, but they don't share the same architecture. So it gives a common base for discussion like through actions, for example, change of perspective. So it is like the so it's just to answer the question Euclidean projective,

we're not going in the direction in terms of research where we're trying to say that you can go from the clean projective for the same agent, we so agents are hard coded with the latent space, which is as some structure. And then we try to exploit it for function for communication for multi agent, like behavior so that they can collaborate. This is it.

Okay, so each agent is within its own projective setting, and then the Euclidean sets the stage and allows that multi perspective.

Yeah, the Euclidean space, which is the outside world, just here in this toy model, because we it's a way to reference the configuration of the world. But for network, this would be, for example, the configuration of its sensors. So it's just here, there's a there's this kind of ambiguity between space and space. So outside space, which is Euclidean and inside, which is projective. But the outside space is just a way to discuss configurations of something that is related to the environment.

So it can be like your your sensors, the configuration of your sensors or something like this information coming from your sensors. So you just give a state a space from it. And the geometry is really in the reconstructed world that is internal. But the being able that you do networks that do this is somewhere that we're working on now. And it's not that obvious because you need to, there's a lot of algorithms, the problem algorithmic problems that we try to solve and that we need to solve. So in this toy model is really like the ambiguity comes only from the fact that it's a toy model. And but what is space is inside it, what is reconstructed is the thing that we perceive space from sensors as being a whole as being something that structures the information.

And so David wants to say something else. Well, I mean, yes, indeed, in the talk that you are referring to, I was giving and also, Kenneth Relieford, if I remember well, yeah, indeed, the toy models encoded a certain world model in an Euclidean way. And then using homogeneous coordinate and this kind of stuff to do the projective

transformation, we just transform the Euclidean space into a projective space.

And you can go, of course, it's in, you can invert it. I mean, there is an online division, but you can invert it.

Anyway, so you can go from one to the other. But Grégoire is right that it was a choice of modeling.

It was a bit motivated by my experience as a brain scientist. You know, there are reasons to believe that memory,

for instance, place cells and grid cells that then, though there are new hypotheses about their hyperbolic characters, but they, you know, usually they are thought as encoding space in an Euclidean manner.

So since when we project, you know, future action, we use sometimes and we have to use memory or even if we

do a remembrance of the past and we project yourself in the scenes in the past, there is this access to memory systems that, you know, we used to think and probably this is still the case on code information in an Euclidean manner. So it would make sense that then for the conscious access, there would be some operation that would allow us to go from Euclidean to projective, which, you know, was implemented that way. At the same time, but this is way beyond my, my abilities in math. So Grégoire will correct me.

There is a way of seeing projective geometry as being more general, as an extension of a fine space by

adding

point at infinity. So if now you think about the projective space with a metric, basically the projective space is an extension of Euclidean space. Am I saying something completely false?

No, I mean, it's one way to seeing it. And there are also a lot of operations that use dialogues, I would say, or exchange between projective and Euclidean. If you think about multi-view reconstruction of 3D Euclidean space from multiple shots in your camera, you take several shots of a building from different perspectives. And then you can use a deterministic algorithm, I mean, approach using epipolar geometry to kind of infer under some priors that basically all lines that converge at point at infinity are actually parallel. Then you reconstruct, you go from projective to Euclidean. So there is a, you know, a deep relationship between the two that is possible and probably functional.

Yeah, that's, that's about it. So, and epipolar geometry is a, is a subset of projective geometry.

So, yeah, so just to give some context.

David has a better intuition on projective spaces than me. This is like, so when he says stuff, I mean, I, I, I, I, like, I, I know he really has a very, uh, I mean, this is crazy, like, a very intuitive, like physical, personal, like visualization. And most of the time he's right. So, are we experiencing a projective geometry or what is it like to experience a projective geometry?

So when, one problem that there's this, with this model is the fact that you have, so you need to force the point at infinity to be, um, so the plane you take away from the projected face space has to be behind. And so you, you're not really working with real maps when, in the, uh, no, you can always extend them. That's okay. But I mean, the thing is, if you want to see it in the way that we see it, us, which is, we have something in front of us, I find it a bit restrictive to say that you cannot see what's behind you. I mean, you never see it, right?

I mean, you can imagine it, but which is not the same thing because you change your frame. So I can't answer really this question. I'm just pointing out the fact that for me, it's very perturbing to think that I have a plane that corresponds to, uh, this continuity, uh, I mean, with respect to, I mean, I have a plane that destroys completely the way that I can think about movement behind me, which is that if I had to imagine, I had a whole space in that is around me and not just in front of me, things would be very, very weird in terms of transformation.

But it's not something that is not possible. I mean, it's, uh, this is what happens when you take psychedelic. You basically, uh, allows the projective transformation to have all its degrees of freedom, but in the perception, like in sensory motor processes, this is calibrated. This is restricted.

So we have to choose certain subset of the group, basically. It's only certain projective transformation that will be used in practice. But now if you go into mystical experience or you take drugs, you're going to mess precisely. That's the hypothesis with those, with those parameters.

And now you have the full fledged 15 degrees of freedom transformation. And you see things that are very weird, like that looks like mandalas or things like that. So that's something you can get when you, when you leave too much freedom to the projective action.

What is it? What if what we're seeing quote in front of us is behind us or something like that?

What do you mean?

What was the, what was the issue with not being able to see behind us?

Yeah. The issue is, so, uh, if you want to see the project, this space as a 3d space, as a way that we see it now, you need to take away a plane and basically projective transformation. So there's a sort of plane at infinity on which you can, I mean, it doesn't exist in the projective space. It exists only in the way you define an Euclidian chart, like a 3d, like a R^3 chart, uh, with the plane that is taken away. And so this plane that is behind you, if you apply a projective transformation, like in front of you, everything will seem fine. You're, you're expecting like it's things that you experience, but then once what happened is that it can send things at infinity back behind you. And so this is something that is a bit, I mean, for me, I'm very on, on, I mean, it's if I, so you can write it explicitly, there's no problem mathematically, but if it's really the way that we experience it, it's extremely weird to think about. So I prefer not thinking about it. So this is the only thing I, and I don't take drugs. So I leave it to other people. I mean, it's, it's really, it could, it's true that taking drugs in this context makes complete, like you can, you could model it in the projective framework and not in the Euclidean framework, because you have a huge amount of projective transformations that make sense more than the one that are related to actions in there. So in the presentation I gave, there was a way to relate that like Euclidean frames to protected frames under certain axioms. And in fact, you have a huge, much bigger space of projective transformations than the one that are restricted through by, uh, the actions of the agents. So the action of the agents inside of the real space. So the Euclidean space induced projective transformation in its internal world, but there are much more projective transformation than these actions. And so you can imagine in this setting, having like very confusing states, uh, of your mind in a way, which you cannot in the opinion case. So I wonder if, if a creature with, or a robot with 360 degree cameras, would it not be so perplexed as it wouldn't necessarily have a visual before or a visual in front and behind? And the thing is, how do you glue them together?

Hmm. You still have to pick a point. You still have to pick a point of convergence, right?

I mean, you need to, you need to find a way to do multimodal integration. Like if you, if you want to have a camera, that's 360, uh, you have, you have the space to represent it. So maybe you can represent it as a sphere. And in this case, the sensors, the sensors are different. So it's, it's not necessarily the same, the same robot than the models that we're considering, but there might be also another homogeneous space. So a G space, it's a space acting like, which corresponds to the rotations of the, uh, of, uh, of, uh, like if you say that it can look 360 and then you can rotate and the space of observation stays the same. So you can always imagine acting on it. And so you can imagine like having another G space structure. And the good thing about it is, is just with respect to the perspective. If you have an agent that can take a different perspective on this environment, which depends on its sensors and its actions and the kind of data you have, then you can define G space and you can do exactly the same setting, everything like it's, it's, it's made in the moral general, more general context, so you can apply it. And so it's not limited only to the projective case. I,

like we come from the PCM projective consciousness model, but, uh, I mean the, the framework now, or at least the way to replace this framework inside of, uh, uh, optimal control is, uh, well adapted for any change of perspective and not in certain, not just projective.

Awesome. Well, where are you gonna go to next? Where will the epistemic drive take you carrying forward?

So there are two projects now. So the first one is to be able to be like, have a, maybe David can talk, uh, but it's to be able to, uh, to get, uh, so to implement it so that it can be used, uh, for, uh, monitoring behaviors or to be able to predict behaviors or to analyze it. So you have model of behaviors of agents like maladaptive behaviors, think that we already published, but then more limited context now to make it inside of the computational framework so that we can use it to analyze like experiments with humans. So this is the first project, which is ongoing with several students. So on the page that I put on my, on the first slide, there's all the work that we're doing, the people are working with us and we're very grateful for them working with us and all the ongoing work in this direction. And there's the second one, which is more like, uh, say, uh, machine learning. And so for me, it's very important to have effective algorithms, things, I mean, like things that you can really use in practice, things that you can really implement and implementing when you have group structures, there are many, many problems that appear. How do you do stochastic optimal control when the latent space is a homogeneous space is, I think a completely open question. And I start to have answers in this direction. So I hope, hopefully the, the, the work will go out to come out soon. And let's say, uh, January or something like this, but it's, there's a huge work in this direction. Cool. Anything else you want to add?

Thank you very much for the invitation.

A lot to think about slash mentally rotate slash project. I mean, after all, as you even alluded to with communication, this is a perspective taking question. Technology is adding some new flavors and methods, for example, asynchronous communication, but it's synchronous for the agent when they perceive it,

but all these different modalities of communication. And it's quite interesting to think about and just really cool that you and colleagues are pursuing that from empirical data analytic and from theoretical mathematical approaches. I mean, on this side for communication, I think, and so it's something I find extremely interesting. Like more generally, if some people want to, some people want to work on the project, please tell us, and I will be very happy. And if you, and even if you have money, please tell us and who we're very happy. So I agree. Great. Great. We were very open. You can always send us a mail. We'll be very happy to discuss about this. Cool. Okay. Well, thank you again to, to, um, all till next time. Okay. See ya. Bye. Bye. Bye. Bye.

Bye.

Bye.